Large Language Models for Generative Recommendation and User Behavior Modeling: Paradigms, Architectures, and Frontiers

1 Introduction

The landscape of recommender systems is undergoing a seismic shift, transitioning from traditional discriminative frameworks that rank predefined candidates to generative paradigms that synthesize recommendations as natural language sequences. Historically, recommendation engines have relied on collaborative filtering and deep learning architectures designed to predict interaction scores or classify user preferences within a fixed item catalog [1; 2]. While models like DeepCoNN successfully leveraged review text to learn latent user and item factors [3], and session-based approaches utilized recurrent neural networks to capture sequential dynamics [4], these methods fundamentally treat recommendation as a retrieval and ranking problem. They often struggle with data sparsity, cold-start scenarios, and the inability to reason over complex, open-world knowledge, limitations that are increasingly apparent in modern, heterogeneous information environments [5].

The emergence of Large Language Models (LLMs) has catalyzed a conceptual reformation, recasting recommendation as a generative task akin to natural language processing. Unlike conventional models that decompose higher-order interactions into pairwise relationships [6], LLMs possess inherent capabilities for semantic understanding, logical reasoning, and context-aware generation [7]. This paradigm shift enables the direct generation of item identifiers, descriptive rationales, or even entire behavioral sequences, effectively unifying user modeling and item synthesis within a single autoregressive framework [8]. By treating user history and item metadata as textual prompts, LLMs can infer latent intents and psychological profiles that transcend surface-level interaction patterns, addressing long-standing challenges in interpretability and personalization [9].

The motivation for this transition is rooted in the unique advantages LLMs offer over discriminative baselines. First, the extensive world knowledge encoded during pre-training allows LLMs to generalize to unseen items and domains, mitigating the cold-start problem that plagues ID-based models [10]. Second, the reasoning capabilities of LLMs facilitate causal inference and counterfactual simulation, enabling systems to distinguish between spurious correlations and genuine user preferences [11]. Third, the generative nature of these models supports the synthesis of natural language explanations, bridging the gap between black-box algorithmic decisions and human-interpretable logic [12]. However, this transition is not without challenges; issues such as hallucination of non-existent items, computational inefficiency in decoding long sequences, and the alignment of generative outputs with specific user constraints remain critical areas of investigation [13; 14].

This survey provides a comprehensive taxonomy of the architectural frameworks, behavioral modeling techniques, and alignment strategies defining this new frontier. We systematically examine semantic item indexing strategies that map discrete catalogs to continuous semantic spaces, parameter-efficient adaptation mechanisms like LoRA for personalized fine-tuning, and hybrid neuro-symbolic architectures that fuse collaborative signals with semantic reasoning [15; 16]. Furthermore, we explore advanced evaluation protocols that move beyond traditional ranking metrics to assess semantic coherence, explanation quality, and long-term user engagement. By synthesizing current advancements and identifying open challenges in scalability, privacy, and ethical governance, this work aims to establish a foundational roadmap for the next generation of generative recommendation systems, guiding the community toward robust, trustworthy, and truly intelligent user modeling solutions [17; 18].

2 Architectural Frameworks for Generative Recommendation

2.1 Semantic Item Indexing and Tokenization Paradigms

The transition from discriminative ranking to generative recommendation necessitates a fundamental reimagining of item representation, where the core challenge lies in mapping discrete, high-cardinality catalogs into the continuous semantic space of Large Language Models (LLMs). Traditional ID-based embeddings, while effective for collaborative filtering, are opaque to LLMs and fail to leverage pre-trained world knowledge. Consequently, recent paradigms focus on constructing semantic identifiers that balance unique item distinction with rich contextual meaning. A primary strategy involves learnable collaborative tokenization, which transforms latent collaborative signals into discrete tokens compatible with LLM vocabularies. For instance, [19] proposes a Masked Vector-Quantized tokenizer that quantizes collaborative filtering representations into discrete tokens, effectively embedding high-order interaction patterns directly into the generative process. Similarly, [20] integrates Residual Quantized VAEs with contrastive alignment to ensure that generated identifiers preserve both semantic hierarchy and collaborative similarity, mitigating the code assignment bias often found in naive discretization.

Complementing collaborative approaches, hybrid indexing schemes synthesize numeric IDs, textual attributes, and category hierarchies to enhance generative flexibility. [21] pioneered the use of Semantic IDs, where items are represented as tuples of codewords derived from quantized item features, enabling the model to generalize to unseen items through semantic reasoning. This concept is further advanced by [22], which employs a hierarchical Graph RQ-VAE to assign multimodal Rec-IDs, fusing visual and collaborative information into a unified token sequence. Furthermore, [23] systematically evaluates indexing strategies, demonstrating that hybrid methods combining sequential, collaborative, and semantic signals significantly outperform trivial random or title-based indexing, particularly in cold-start scenarios. To bridge the gap between continuous collaborative embeddings and discrete text tokens, [24] introduces a text-like encoding mechanism that converts collaborative embeddings into binary sequences, allowing LLMs to process interaction data natively without architectural modification.

However, expanding the vocabulary to accommodate item-specific tokens introduces the critical risk of hallucination, where models generate non-existent items. To address this, constrained decoding and vocabulary grounding mechanisms are essential. [25] proposes a two-step framework where the LLM first generates meaningful tokens that are subsequently mapped to valid items via a grounding layer, ensuring factual consistency. Additionally, [26] highlights how standard decoding strategies can amplify bias toward “ghost tokens,” advocating for debiasing techniques that enforce diversity and validity. The integration of trie-based decoding or specialized vocabulary pruning, as discussed in the context of generative retrieval, further restricts the output space to the actual item corpus. Looking forward, the field is moving toward dynamic, context-aware tokenization that adapts to user intent in real-time, potentially leveraging virtual tokens as suggested by [27] to decouple item representation from fixed vocabularies, thereby enhancing scalability and reducing the parameter overhead associated with massive item catalogs.

2.2 Parameter-Efficient Personalization and Adaptation Strategies

While the preceding section established semantic identifiers as the foundational layer for mapping items into the LLM space, the subsequent realization of generative recommendation requires robust strategies to adapt these massive pre-trained models to dynamic, high-cardinality user behaviors. Full parameter fine-tuning is often prohibitive due to the sheer scale of modern LLMs and the necessity of maintaining distinct user profiles. Consequently, Parameter-Efficient Fine-Tuning (PEFT) has emerged as the dominant paradigm, enabling the injection of user-specific signals into frozen backbones while preserving general world knowledge. A primary approach involves Low-Rank Adaptation (LoRA), where lightweight modules are injected into transformer layers to capture domain-specific or user-specific patterns. Recent advancements propose maintaining independent LoRA modules for individual users or clusters, effectively decoupling personalized behavior modeling from the foundational model weights [15]. This “One PEFT Per User” (OPPU) strategy allows for scalable personalization where user parameters can be stored and retrieved on demand, significantly reducing storage costs compared to full model replication while outperforming prompt-based methods in capturing complex behavioral shifts [28]. Furthermore, frameworks like [29] extend this concept to black-box settings by factorizing the model into a shared base and multiple user-specific heads, leveraging rerankers to prioritize relevant historical records for adapter training.

Complementary to these weight-based adaptation techniques, prompt tuning and soft prompt optimization offer a non-intrusive mechanism for contextualizing LLMs. By learning continuous vector representations that prefix the input sequence, models can internalize user preferences without altering the underlying parameters. Theoretical analyses suggest that prompt tuning provides robust guarantees for downstream tasks, particularly when the pre-trained model functions as a latent variable generator [30]. In recommendation contexts, this manifests as the [31] paradigm, which unifies diverse tasks through personalized prompts, allowing for zero-shot or few-shot adaptation that drastically reduces the need for extensive retraining [31]. Moreover, retrieval-augmented strategies enhance this by dynamically assembling user history into the context window. Approaches such as [32] address the “lifelong sequential behavior incomprehension” problem by retrieving semantically relevant user behaviors to construct evidence-rich prompts, enabling LLMs to comprehend long-term interests even with limited context windows [32]. Similarly, [33] integrates distilled user embeddings via cross-attention or soft-prompts, offering a computationally efficient alternative to text-heavy prompting for long sequence tasks [33].

A critical challenge in these adaptive architectures is catastrophic forgetting, where adapting to new user data degrades performance on previously learned distributions. To mitigate this, lifelong and incremental adaptation architectures employ dual-module systems or memory-augmented networks. For instance, [34] highlights the necessity of distinguishing between vertical continuity (general to specific) and horizontal continuity (temporal evolution), advocating for mechanisms that isolate task-specific parameters. Memory-augmented approaches, such as those utilizing external memory banks to cache long-term contexts, allow models to update user representations without gradient-based optimization on the entire history [35]. Additionally, preference alignment techniques like Direct Preference Optimization (DPO) have been adapted for personalization, enabling stable alignment with individual user feedback through closed-form solutions that avoid the instability of traditional Reinforcement Learning from Human Feedback (RLHF) [36; 37]. Future directions likely involve hybridizing these strategies, combining the expressiveness of LoRA with the flexibility of retrieval-augmented prompting to achieve real-time, privacy-preserving personalization at web scale. Building upon these parameter-efficient and prompt-based strategies, the next evolution involves integrating LLMs with traditional collaborative filtering and graph-based architectures to explicitly model high-order connectivity patterns that pure language modeling may overlook.

2.3 Hybrid Neuro-Symbolic and Collaborative Fusion Architectures

The integration of Large Language Models (LLMs) with traditional collaborative filtering and graph-based architectures represents a pivotal evolution in generative recommendation, aiming to synthesize the semantic reasoning prowess of foundation models with the structural precision of interaction graphs. While LLMs excel at interpreting unstructured content and inferring latent user intents through natural language, they often lack the explicit modeling of high-order connectivity patterns inherent in user-item bipartite graphs. Conversely, Graph Neural Networks (GNNs) and collaborative filtering models capture complex relational dependencies but struggle with Cold-Start scenarios due to their reliance on ID-based representations and limited world knowledge. Hybrid neuro-symbolic frameworks address this dichotomy by enforcing a bidirectional flow of information: utilizing graph structures to ground LLM hallucinations in observed interaction data, and leveraging LLMs to enrich sparse graph nodes with semantic attributes.

A primary architectural paradigm involves graph-augmented language modeling, where structural signals are injected directly into the LLM's attention mechanisms or prompt contexts. For instance, [38] proposes a framework that employs LLM-based graph augmentation strategies to reinforce user-item interaction edges and enhance node attribute understanding, effectively mitigating data sparsity by translating collaborative signals into natural language prompts. Similarly, [39] introduces a prompt construction framework that explicitly encodes relational graph information into textual expressions, enabling LLMs to intuitively grasp connectivity patterns that are otherwise opaque to pure text-based models. These approaches often utilize GNNs as prefix modules or encoders to generate structural embeddings that are concatenated with textual tokens, a strategy echoed in the broader taxonomy of LLMs-on-graphs where GNNs serve as specialized feature extractors for the language backbone [40].

Beyond simple feature concatenation, advanced fusion mechanisms employ variational inference and mutual information maximization to align the disparate latent spaces of symbolic graph models and neural language models. [41] establishes a theoretical foundation for this alignment, proposing a cross-view framework that maximizes the mutual information between collaborative relational signals and LLM-derived semantic representations. This ensures that the generative process is not only semantically coherent but also statistically consistent with historical interaction patterns. Furthermore, [42] demonstrates the efficacy of mutual augmentation, where an adaptive aggregation module dynamically fuses predictions from conventional collaborative models and LLM-based generators, allowing the system to leverage the strengths of both paradigms based on sample difficulty and data density.

The challenge of scaling these hybrid architectures to industrial magnitudes has spurred innovations in efficiency and modular design. [43] addresses the computational bottleneck of joint training by introducing a tunable side structure that connects frozen LLMs with GNNs, significantly reducing memory overhead while preserving performance on textual graphs. Additionally, the concept of "neuro-symbolic" reasoning is extended through automaton-augmented retrieval, where finite automata derived from graph structures guide the LLM's decoding path, effectively constraining the generation space to valid item transitions [44]. Despite these advancements, critical trade-offs remain regarding the latency of multi-stage fusion pipelines and the potential for signal dilution when merging high-dimensional semantic vectors with structural embeddings. Future research must therefore focus on developing unified foundation models that natively internalize graph topology within their pre-training objectives, moving beyond modular fusion toward end-to-end differentiable architectures capable of simultaneous semantic and structural reasoning.

2.4 Retrieval-Augmented Generative Recommendation Designs

Bridging the gap between the unified foundation models envisioned in hybrid neuro-symbolic frameworks and practical industrial deployment, Retrieval-Augmented Generative Recommendation (RAGR) architectures represent a pivotal evolution in generative paradigms. While previous approaches sought to internalize graph topology within model weights, RAGR addresses the inherent limitations of parametric knowledge in Large Language Models (LLMs)—such as hallucination and static world knowledge—by decoupling the retrieval of candidate items or contextual evidence from the generative decoding process. This separation enhances factual consistency, reduces computational overhead, and enables effective modeling of long-tail distributions. The core principle involves augmenting the LLM's input context with dynamically retrieved external data, thereby grounding the generation in observable user behaviors and up-to-date item catalogs rather than relying solely on internal weights.

A primary design pattern within this domain is Collaborative Retrieval-Augmented Generation, where relevant user-item interaction histories or similar user profiles are retrieved to construct evidence-rich prompts. This approach guides the LLM to align its reasoning with dataset-specific collaborative patterns, mitigating the risk of generating items that contradict established user preferences. For instance, [32] introduces a retrieval-enhanced framework that performs semantic user behavior retrieval to compress long interaction sequences, significantly improving the model's ability to comprehend lifelong sequential behaviors in both zero-shot and few-shot settings. Similarly, [45] optimizes the representation of user data through collaborative refinement, demonstrating that efficient retrieval of personalized documents can substantially reduce context window requirements while maintaining high recommendation accuracy. These methods effectively transform the recommendation task from pure generation to a grounded inference problem, leveraging external memory to overcome the "lost in the middle" phenomenon common in long-context LLMs.

Complementary to collaborative retrieval, Two-Stage Retrieve-then-Generate pipelines have emerged as a robust strategy for balancing efficiency and semantic depth, echoing the modular fusion concepts discussed in graph-augmented architectures. In these systems, a lightweight, often discriminative, retriever first narrows the candidate space based on collaborative signals or semantic similarity, followed by an LLM that reranks, explains, or generates final selections from this constrained set. [46] exemplifies this by employing user-item interactions for initial candidate retrieval before utilizing a fine-tuned LLM for ranking, explicitly instructing the model to select from the provided candidates to prevent hallucination. Furthermore, [47] proposes a hybrid retrieval mechanism within a federated setting, mining ID-based patterns and text-based features before feeding them into an LLM for re-ranking, thereby addressing data heterogeneity and privacy constraints simultaneously. This decoupling allows the system to leverage the scalability of traditional retrieval models while capitalizing on the LLM's superior reasoning and explanation capabilities.

Beyond user history, integrating External Knowledge Bases is critical for injecting real-time facts and domain-specific constraints, further enriching the semantic grounding initiated by graph-based methods. [10] enhances LLM-based recommendations by incorporating external knowledge not only into prompts but also through a knowledge-based contrastive learning scheme, ensuring that generated recommendations are factually grounded and contextually relevant. Similarly, [38] utilizes LLMs to augment interaction graphs with rich content information, effectively bridging the gap between sparse implicit feedback and dense semantic understanding. However, challenges remain in optimizing the retrieval component itself; [48] highlights the necessity of optimizing retrievers via reinforcement learning or knowledge distillation to better align retrieved contexts with downstream generation objectives.

Looking forward, the convergence of generative retrieval and retrieval-augmented generation presents a promising frontier that may eventually realize the end-to-end differentiable systems anticipated in prior sections. While current RAGR designs rely on external retrievers, emerging works like [49] suggest a unified approach where the model learns to generate document identifiers directly, potentially merging the retrieval and generation stages into a single autoregressive process. Nevertheless, for immediate industrial applicability, hybrid architectures that maintain a distinct, optimizable retrieval module offer the most viable path toward scalable, faithful, and personalized generative recommendation systems.

3 Modeling Dynamic User Behaviors and Sequential Patterns

3.1 Semantic Serialization of Heterogeneous Behavioral Sequences

The semantic serialization of heterogeneous behavioral sequences constitutes the foundational preprocessing step for leveraging Large Language Models (LLMs) in generative recommendation, necessitating the transformation of discrete, multi-modal interaction logs into coherent natural language narratives. Traditional sequential models often struggle with the sparsity and modality gaps inherent in raw clickstreams, whereas LLMs require inputs that align with their pre-trained semantic spaces to effectively reason over user intent. A primary strategy involves multi-behavior tokenization, where distinct interaction types—such as clicks, purchases, skips, and dwell times—are encoded into semantic tokens or descriptive sentences that preserve temporal order and contextual nuance. For instance, [50] projects heterogeneous behaviors into multiple latent semantic spaces to facilitate influence modeling via self-attention, while [51] employs behavior-specific transition matrices to capture short-term contexts within a recurrent framework. Extending this, [52] constructs multi-relational graphs to globally encode item-to-item relations across auxiliary behavior types, addressing the limitations of single-behavior sequences.

A critical challenge in this serialization process is the unification of cross-domain sequences, where user activities span disparate domains such as shopping, video streaming, and news consumption. LLMs offer a unique advantage here by leveraging their pre-trained world knowledge to identify latent connections between these disparate activities, effectively merging them into a single chronological narrative. [53] demonstrates the efficacy of unifying user embeddings from different domains into a global representation, while [54] further integrates structural information into LLMs via user retrieval to enhance cross-domain sequential modeling. Furthermore, [55] explicitly extracts and fuses heterogeneous knowledge from diverse user behaviors through instruction tuning, mitigating feature sparsity and knowledge fragmentation.

The optimization of temporal granularity within the LLM's context window remains a pivotal consideration. Balancing absolute timestamps against relative intervals is essential for capturing both immediate session dynamics and long-term evolutionary trends without exceeding computational limits. [56] highlights the necessity of incorporating time gaps and time-of-day information directly into model dynamics to improve next-event prediction. More recently, [57] and [58] advocate for selective state space models to handle lifelong sequences efficiently, addressing the quadratic complexity of attention mechanisms in long-range dependency modeling. Additionally, [59] proposes reprogramming input time series with text prototypes to align numerical temporal sequences with the LLM's reasoning capabilities, a technique increasingly adapted for behavioral forecasting.

Despite these advances, trade-offs persist between the richness of semantic description and the efficiency of inference. While detailed narrative construction enhances interpretability and intent inference, as seen in [60], it introduces significant token overhead. Future research must therefore focus on adaptive serialization strategies that dynamically adjust the level of semantic detail based on the complexity of the user's interest journey, potentially leveraging hierarchical summarization techniques [61] to compress long-term histories while retaining critical semantic signals for generative reasoning.

3.2 Capturing Long-Range Dependencies and Evolving Interests

Building upon the semantic serialization and temporal granularity strategies discussed previously, the focus now shifts to architecturally modeling the non-linear evolution of user preferences over extended time horizons. While prior sections addressed how to format and compress behavioral sequences, a paramount challenge remains in processing these lifelong logs without succumbing to information dilution or the vanishing gradient problem inherent in traditional recurrent architectures. To retain fidelity across sequences spanning thousands of interactions, recent innovations have prioritized decoupling short-term session dynamics from long-term evolutionary trends through hierarchical memory mechanisms and state space models (SSMs). For instance, the Dynamic Memory-based Attention Network (DMAN) segments long behavior sequences into sub-sequences, utilizing memory blocks to preserve long-term interests while explicitly extracting short-term intents for joint recommendation [62]. Similarly, the Hierarchical Periodic Memory Network addresses lifelong sequential modeling by employing a periodical updating mechanism to capture multi-scale patterns, effectively handling the extreme length of user histories accumulated over years [63].

Complementing these memory-centric approaches, recent advancements have integrated SSMs, such as Mamba, to achieve linear complexity scaling, offering a computationally efficient alternative to the quadratic attention mechanisms of Transformers. Models like Mamba4Rec and RecMamba leverage selective state spaces to model lifelong sequences (length $\geq$ 2k), significantly reducing training duration and memory costs while maintaining or surpassing the performance of attention-based baselines [58; 57]. These approaches are further enhanced by hybrid architectures like EchoMamba4Rec, which harmonizes bidirectional SSMs with spectral filtering to capture complex dependencies and reduce noise in user interaction data [64]. Beyond architectural restructuring, dynamic interest decomposition has emerged as a critical strategy for disentangling static preferences from transient interests. The Multi-temporal-range Mixture Model (M3) employs a learned gating mechanism to adaptively combine models with different temporal ranges, addressing the co-existence of short-term and long-term dependent behavioral patterns [65]. Furthermore, consistency-aware gating networks dynamically adapt to preference evolution by balancing general user preferences with current priority models, ensuring recommendations remain relevant as user interests drift [66].

To further refine the modeling of preference evolution, the integration of continuous-time encoding schemes moves beyond discrete step models to capture the precise duration between interactions. Time-aware architectures, such as the Time-Aware Recurrent Log-BiLinear (TA-RLBL) model and the Multi-hop Time-aware Attentive Memory network (MTAM), incorporate temporal gating methodologies to model the decay of interest and the impact of time gaps on subsequent actions [51; 67]. These models parametrize hidden unit transitions as a function of context information, allowing for a more nuanced understanding of user state evolution [56]. Looking forward, the convergence of retrieval-enhanced frameworks with SSMs offers a promising trajectory to address the "lifelong sequential behavior incomprehension" problem. Approaches like ReLLa retrieve semantically relevant behaviors to augment LLM contexts, thereby improving zero-shot and few-shot recommendation capabilities without requiring exhaustive sequence processing [32]. Similarly, Behavior Aggregated Hierarchical Encoding (BAHE) decouples atomic behavior encoding from inter-behavior interactions to break the efficiency barriers of LLMs in processing billion-scale user sequences [68]. By unifying these disparate temporal modeling paradigms into cohesive frameworks that achieve linear scalability and continuous-time precision, the field establishes the necessary structural foundation to transition from modeling *when* interactions occur to inferring *why* they happen, setting the stage for the latent intent inference techniques explored in the subsequent section.

3.3 Latent Intent Inference and Psychological Profiling

Moving beyond the superficial modeling of interaction sequences, latent intent inference leverages the profound reasoning capabilities of Large Language Models (LLMs) to deduce unobserved user motivations, psychological dispositions, and high-level "interest journeys" from sparse behavioral signals. Traditional approaches often rely on aggregated feature extraction or hidden state compression, which may fail to preserve the integrity of specific highly correlated behaviors [50]. In contrast, LLMs enable a paradigm shift towards deep semantic understanding, treating user history not merely as a sequence of IDs but as a narrative rich with implicit psychological cues. This allows for the extraction of latent traits, such as Big Five personality dimensions, directly from review histories and interaction patterns, facilitating hyper-personalized strategies aligned with user psychometrics [69].

A critical advancement in this domain is the ability to perform zero-shot personality and trait extraction. By prompting LLMs to analyze the linguistic style and content of user-generated text, systems can infer stable psychological dispositions that govern long-term preferences, overcoming the data sparsity issues that plague conventional collaborative filtering [70]. Furthermore, LLMs excel at summarizing and abstracting a user's "interest journey," identifying persistent themes and evolving hobbies that transcend individual item interactions. Frameworks like [61] demonstrate how hierarchical and recurrent summarization techniques can compress extensive behavioral logs into coherent preference profiles, enabling the model to capture the non-linear evolution of user interests over extended time horizons. This abstraction is crucial for distinguishing between transient novelty seeking and genuine shifts in core preferences, a nuance often lost in discrete step models [71].

Moreover, LLMs provide sophisticated mechanisms for interpreting implicit feedback through sentiment analysis and intent disambiguation. Unlike traditional models that treat a "skip" or "pause" as a uniform negative signal, LLMs can contextualize these actions to distinguish between disinterest, prior consumption, or temporary distraction. This capability is enhanced by frameworks that integrate causal discovery to assess personalized effects, as seen in nutrition-oriented recommenders where latent health goals are inferred from dietary choices [72]. The synergy between LLM reasoning and structural learning is further exemplified by approaches that combine LLMs with hypergraph neural networks to explicitly account for the intricacies of human preferences, thereby enhancing explainability and recommendation accuracy [73].

However, challenges remain in ensuring the fidelity of these inferred profiles. Research indicates that LLMs may exhibit "sycophancy" or align too closely with superficial prompt structures, potentially distorting the true latent intent [74]. Additionally, the separability of concept variables in text remains limited, suggesting that distilling specific latent traits without losing semantic context is non-trivial [75]. Future research must therefore focus on developing robust alignment strategies, such as learning latent preferences from user edits [76], to ensure that psychological profiling remains grounded in authentic user behavior rather than model hallucinations. By bridging the gap between surface-level patterns and deep motivational structures, latent intent inference positions LLMs as pivotal tools for constructing truly human-centric recommendation ecosystems.

3.4 Causal Reasoning and Counterfactual Behavior Simulation

Building on the foundation of latent intent inference, which uncovers the psychological motivations behind user actions, the next logical step is to distinguish between mere correlation and genuine causation. The integration of causal inference frameworks with Large Language Models (LLMs) marks a paradigmatic shift from correlational pattern matching to mechanistic understanding in user behavior modeling. Traditional sequential recommenders often conflate spurious correlations with genuine preferences, particularly when confounded by popularity bias or position effects. LLMs, endowed with extensive world knowledge and reasoning capabilities, offer a unique avenue to disentangle these factors by constructing structural causal models (SCMs) from observational data. Recent advances demonstrate that LLMs can act as virtual domain experts to infer causal orders and construct causal graphs, significantly improving the accuracy of causal discovery compared to purely statistical methods [77; 78]. By leveraging metadata and natural language descriptions, frameworks like [79] synergize data-driven algorithms with LLM reasoning to automate the generation of resilient causal graphs, thereby enabling more robust policy learning.

A critical application of this causal perspective is counterfactual behavior simulation, which addresses the inherent sparsity and selection bias in interaction logs. The fundamental counterfactual query—"what would the user's decision be if the recommended items had been different?"—allows for the generation of synthetic training data that mimics interventions not present in the historical record [80]. This approach reformulates recommendation as a causal inference problem, utilizing structural equation models optimized on observed data to simulate user feedback under hypothetical ranking scenarios. Furthermore, LLMs facilitate the creation of generative agents capable of simulating complex, emotion-driven user trajectories in response to such interventions, providing a sandbox for evaluating long-term reward policies without risking real-world user experience [81; 82]. These agents, equipped with memory and reflection mechanisms, can emulate the filter bubble effect and uncover underlying causal relationships that static models miss.

However, the fidelity of LLM-generated counterfactuals remains a subject of rigorous scrutiny. While LLMs produce fluent counterfactual explanations, they often struggle to maintain minimality in input perturbations, potentially introducing artifacts that diverge from human behavioral logic [83]. The "artifactuality" of synthetic data poses risks of model collapse or bias amplification if not carefully regulated [84]. To mitigate this, recent works propose hybrid architectures where LLMs guide causal discovery but are constrained by formal causal criteria or verified against held-out empirical distributions [85]. For instance, in personalized nutrition systems, causal personal models estimated via LLMs are combined with population-level data to ensure that counterfactual recommendations adhere to biological constraints [72].

Looking forward, the convergence of causal inference and generative modeling promises to resolve the disconnect between offline metrics and online performance. By treating LLMs as simulacra for counterfactual evaluation, researchers can optimize for long-term user sustainability rather than immediate click-through rates. Future research must focus on developing rigorous validation protocols to ensure that simulated counterfactuals preserve the stochastic nature of human decision-making and do not merely reflect the priors embedded in the pre-trained language model. The ultimate goal is a self-correcting recommendation ecosystem where causal reasoning guides the generation of diverse, unbiased, and truly personalized user experiences.

4 Generative Strategies for Personalization and Interaction

4.1 Decoding Mechanisms and Task Unification for Generative Recommendation

The operational efficacy of generative recommendation systems hinges critically on the decoding mechanisms employed to translate latent semantic representations into actionable item sequences, alongside the architectural capacity to unify disparate recommendation objectives. Unlike traditional discriminative models that compute interaction scores over a fixed candidate set, generative paradigms treat recommendation as a sequence-to-sequence generation task, fundamentally altering the inference landscape. The most prevalent approach involves autoregressive item sequence generation, where models predict the next item token conditioned on historical interactions and previously generated tokens. [8] demonstrates that reformulating recommendation as a sequential transduction task allows models to scale effectively with compute, achieving significant performance gains in industrial settings. However, standard autoregressive decoding faces inherent challenges, including amplification bias where length normalization inflates scores for items containing high-probability "ghost tokens," and homogeneity issues leading to repetitive recommendations [26]. To mitigate these, recent innovations such as Debiasing-Diversifying Decoding (D3) disable length normalization for specific tokens and utilize assistant models to encourage diversity, thereby enhancing both accuracy and coverage.

Contrasting with pure autoregressive generation, hybrid strategies have emerged to balance semantic richness with computational efficiency. Verbalizer-based ranking and list-wise optimization decouple candidate retrieval from the generative process, employing verbalizers to map output logits onto probability distributions over pre-retrieved candidate sets. [46] exemplifies this by fine-tuning large language models to rank retrieval candidates provided in natural language format, effectively leveraging the model's reasoning capabilities while constraining the output space to valid items. Similarly, [86] proposes bypassing beam search decoding in favor of a direct item projection head, arguing that the computational overhead of autoregressive search is often unnecessary for sequential tasks, thus achieving substantial efficiency improvements without sacrificing performance. Furthermore, non-autoregressive parallel generation techniques are gaining traction; [87] introduces methods where models learn to plan their generation process hierarchically, enabling parallel token production that significantly reduces latency and key-value cache consumption compared to traditional step-wise decoding.

Beyond decoding mechanics, a transformative trend is the unification of heterogeneous recommendation tasks through instruction following. This paradigm posits that diverse objectives—ranging from rating prediction and sequential recommendation to review summarization and explanation generation—can be formulated as natural language instruction-following problems within a single model architecture. [88] validates this by designing generalized instruction formats that allow models to dynamically switch operational modes based on user prompts, outperforming specialized baselines including GPT-3.5. This unification is further advanced by the "Pretrain, Personalized Prompt, and Predict" (P5) paradigm, which converts all data modalities into natural language sequences, enabling zero-shot or few-shot adaptation across tasks [31]. Moreover, [89] extends this capability by introducing compositional instructions, where complex recommendation workflows are decomposed into chains of subtasks, enhancing the model's ability to handle multi-step reasoning.

Looking forward, the convergence of efficient decoding strategies and unified instruction tuning points toward a future where generative recommenders operate as versatile, agentic systems. The challenge remains to further optimize the trade-off between the semantic flexibility of open-ended generation and the strict constraints required for factual validity and low-latency serving in real-world applications. Integrating constrained decoding with advanced instruction tuning will be pivotal in realizing robust, universal recommendation engines capable of navigating complex user intent landscapes.

4.2 Synthesis of Natural Language Explanations and Rationales

Building upon the unified, instruction-based architectures discussed previously, the synthesis of natural language explanations represents a pivotal advancement in generative recommendation, transitioning systems from opaque scoring mechanisms to transparent, reasoning-capable agents. Unlike traditional post-hoc explainability methods that approximate black-box decisions, Large Language Models (LLMs) enable the direct generation of coherent rationales that articulate the logical nexus between user history and suggested items. This capability is foundational to frameworks like [90], which augment recommender systems by converting user profiles into prompts that facilitate in-context learning, thereby rendering the recommendation process inherently interactive and explainable. Similarly, [88] demonstrates that framing user preferences as natural language instructions allows LLMs to execute recommendation tasks with explicit justifications, effectively bridging the semantic gap between collaborative signals and human-interpretable reasoning.

A critical methodological trend involves the integration of logic-scaffolding and Chain-of-Thought (CoT) reasoning to enhance the faithfulness of generated explanations. By inducing intermediate reasoning steps, models can explicitly trace the derivation of a recommendation before producing the final justification. The efficacy of CoT paradigms in improving accuracy and reasoning depth has been empirically validated in contexts such as [91], where CoT strategies yielded superior performance compared to simple prompting. Furthermore, [92] proposes distilling step-by-step reasoning capabilities from large teacher models into compact student models, ensuring that resource-constrained systems can still generate meaningful, logical rationales without prohibitive computational costs. This approach addresses the challenge of maintaining explanation quality while ensuring scalability, a necessary trade-off for industrial deployment.

To align explanations with specific user needs, personalized aspect-controlled generation has emerged as a vital strategy. Rather than providing generic justifications, systems can condition the tone, focus, and depth of rationales on individual user profiles. [93] highlights that personalized fine-tuning significantly improves model reasoning on subjective tasks, suggesting that explanation generators must be adapted to individual psychological dispositions to be truly effective. Additionally, [73] synergizes LLM reasoning with hypergraph structures to capture multifaceted human interests, enabling explanations that reflect complex, non-linear preference dynamics rather than superficial attribute matching. This human-centric modeling ensures that rationales resonate with the nuanced evolution of user interests over time.

However, the generative nature of these explanations introduces significant challenges regarding faithfulness and hallucination. There is a persistent risk that models may fabricate user preferences or item features to construct plausible-sounding but factually incorrect narratives. [10] addresses this by incorporating external knowledge and contrastive learning schemes to ground LLM outputs, thereby mitigating the reliability issues inherent in purely parametric generation. Moreover, [94] underscores the difficulty in verifying whether generated statements are fully supported by cited references, proposing automated evaluation frameworks to detect attribution errors. Ensuring that explanations are not only fluent but also strictly aligned with the underlying decision logic remains an open problem. Future research must therefore focus on developing constrained decoding strategies and rigorous verification protocols, such as those explored in [95], to guarantee that the explanatory power of LLMs does not come at the expense of factual integrity. These efforts to establish transparent reasoning and faithful explanations provide the essential foundation for the next evolutionary step: shifting from static explanation generation to dynamic, multi-turn conversational systems and autonomous agents capable of proactive engagement.

4.3 Conversational Interfaces and Agentic Recommendation Paradigms

The paradigm of generative recommendation is undergoing a profound transformation, shifting from static, one-off list generation to dynamic, multi-turn conversational systems and autonomous agents capable of proactive engagement. This evolution addresses the inherent limitations of passive recommendation by enabling intent clarification, iterative preference elicitation, and complex reasoning in open-world scenarios. Central to this shift is the capability of Large Language Models (LLMs) to function not merely as rankers but as interactive dialogue managers that can synthesize user history with real-time feedback. Early unified frameworks, such as [96], demonstrated the efficacy of integrating recommendation modules directly into dialog generation via vocabulary pointers, overcoming the disconnect between retrieval and response synthesis found in traditional modular architectures. Furthermore, the zero-shot capabilities of foundational models have been leveraged to convert user profiles into prompts, allowing systems like [90] to establish robust user-item connections through in-context learning without extensive fine-tuning.

A critical challenge in conversational recommendation remains the efficient elicitation of latent user preferences, particularly in cold-start scenarios where interaction data is sparse. Recent advancements propose decision-theoretic approaches that frame preference elicitation as a Bayesian optimization problem. For instance, [97] introduces frameworks that balance exploration and exploitation by utilizing Natural Language Inference to update belief states, significantly outperforming monolithic prompting strategies in identifying top preferences within limited dialogue turns. Complementing this, [98] provides empirical evidence that LLMs can rival fine-tuned baselines in "in-the-wild" conversations, suggesting that the semantic reasoning inherent in pre-trained models serves as a powerful prior for understanding nuanced user intents expressed in natural language.

Beyond reactive dialogue, the frontier of agentic recommendation involves the deployment of autonomous agents capable of planning, tool invocation, and self-reflection. Systems such as [99] exemplify this transition by employing self-inspiring algorithms that allow the agent to iteratively refine its planning path based on historical states, effectively mimicking human-like deliberation. Similarly, [100] posits that LLMs should act as surrogate users that actively invoke external retrieval and ranking tools to probe attribute granularities, thereby bridging the gap between semantic intent and precise item matching. This agentic capability extends to simulation environments, where generative agents are utilized to stress-test recommendation policies. As highlighted in [101], LLM-driven simulators can emulate diverse user behaviors—including feedback loops and preference shifts—providing a scalable alternative to costly human-in-the-loop evaluations. However, the fidelity of these simulations remains a subject of scrutiny, with studies like [84] warning of potential artifacts where synthetic behaviors diverge from genuine human cognitive processes.

Looking forward, the integration of agentic workflows necessitates rigorous alignment strategies to ensure agents adhere to user constraints and ethical norms. The emergence of persona-based alignment, as explored in [102], suggests that dynamic system messages can steer agent behavior toward individualized value systems without retraining. Nevertheless, challenges persist regarding the stability of long-horizon planning and the mitigation of hallucination in open-ended generation. Future research must therefore focus on hybrid neuro-symbolic architectures that ground agentic reasoning in verifiable knowledge graphs while maintaining the flexibility of natural language interaction, ultimately realizing a vision where recommendation systems evolve from passive filters to active, collaborative partners in user decision-making.

4.4 Controllability and Personalized Content Generation

Building upon the agentic workflows and proactive engagement strategies discussed previously, the paradigm of generative recommendation is undergoing a further transformation: shifting from merely retrieving or ranking existing items to synthesizing entirely new, personalized content and strictly controlling output characteristics to align with nuanced user constraints. This evolution addresses the limitations of static catalogs by empowering Large Language Models (LLMs) to function not just as interactive dialogue managers, but as controllable content creators capable of crafting unique user experiences. A primary frontier in this domain is personalized multimodal content synthesis, where LLMs serve as the cognitive engine to condition generative models across diverse modalities. By leveraging discrete sequence modeling, frameworks like [103] demonstrate the feasibility of unifying text, image, audio, and speech within a single LLM architecture, enabling the generation of holistic user experiences. Similarly, [22] extends this capability to recommendation-specific contexts by employing hierarchical quantization to map multimodal item features into semantic tokens, allowing the model to autoregressively generate identifiers that encapsulate both collaborative signals and rich content semantics. Furthermore, the application of LLMs in procedural content generation, as exemplified by [104], illustrates how text prompts can guide the creation of open-ended, diverse environments that reflect specific user play-style dynamics, marking a significant departure from static catalog browsing.

Achieving such high-fidelity personalization, however, requires rigorous mechanisms for constraint-guided decoding to ensure safety, relevance, and adherence to hard user specifications. Traditional unconstrained generation often suffers from hallucinations or violations of budgetary and topical limits, which can undermine the trust established in the conversational phases described earlier. To mitigate this, gradient-based constrained sampling methods, such as MuCoLa proposed in [105], integrate differentiable constraints directly into the energy function of the generation process, enabling non-autoregressive sampling that satisfies complex user-defined rules without sacrificing fluency. Complementing this, [106] introduces DOMINO, a subword-aligned decoding algorithm that enforces strict formal language constraints with negligible computational overhead, addressing the latency barriers often associated with controlled generation in real-time recommendation systems. These approaches are critical for operationalizing generative recommenders in high-stakes domains where factual accuracy and constraint satisfaction are paramount.

Beyond structural and factual constraints, the ability to dynamically adapt the linguistic style and tone of recommendations is essential for fostering empathetic user interactions that resonate with individual personas. Recent advances in activation engineering offer a parameter-efficient alternative to full fine-tuning for style control. [107] demonstrates that adding computed style vectors to hidden layer activations can precisely modulate sentiment and writing style, allowing the system to tailor explanations to individual user personas. This concept is further refined in [108], which utilizes bi-directional preference optimization to derive steering vectors that offer granular control over model behavior, effectively managing trade-offs between truthfulness and creativity. Moreover, [26] highlights the necessity of specialized decoding strategies like Debiasing-Diversifying Decoding (D3) to counteract the homogeneity inherent in standard autoregressive processes, ensuring that personalized content remains diverse and serendipitous rather than repetitive.

Looking forward, the integration of causal reasoning and user edit feedback into the generative loop presents a promising direction for enhancing controllability and closing the gap between agent planning and content realization. Frameworks like [76] suggest that inferring latent preferences from user modifications can create a self-correcting generation policy that evolves with user intent, bridging the gap between the autonomous agents discussed previously and the final synthesized output. As generative models become more capable of synthesizing content indistinguishable from human creation, as noted in [109], the challenge will shift towards maintaining authentic personalization while preventing the erosion of unique user identity through over-optimization. The convergence of these controllability mechanisms with multimodal synthesis capabilities promises to redefine the boundary between recommendation and creation, ushering in an era of truly adaptive, user-centric generative ecosystems.

5 Alignment, Optimization, and Trustworthiness

5.1 Advanced Preference Alignment and Optimization Strategies

The alignment of Large Language Models (LLMs) for generative recommendation has evolved from complex Reinforcement Learning from Human Feedback (RLHF) pipelines toward more stable and computationally efficient preference optimization paradigms. Traditional RLHF, which involves training a separate reward model followed by policy optimization via Proximal Policy Optimization (PPO), often suffers from instability and high computational overhead in recommendation contexts where user preferences are dynamic and multi-faceted [110]. To address these limitations, Direct Preference Optimization (DPO) has emerged as a pivotal alternative, reformulating the alignment objective to directly optimize the policy using a classification loss on preference pairs, thereby eliminating the need for explicit reward modeling and sampling during fine-tuning [36]. This approach has been theoretically grounded as a form of inverse Q-learning at the token level, enabling fine-grained credit assignment that is crucial for sequential recommendation tasks [111].

However, standard DPO assumes a monolithic human preference distribution, which is inadequate for personalized recommendation systems where user tastes are diverse and often conflicting. Recent advancements introduce Multi-Objective and Personalized Alignment frameworks, such as Sequential Preference Optimization (SPO), which aligns models across multiple dimensions (e.g., accuracy, diversity, and novelty) without explicit reward modeling, effectively managing the trade-offs inherent in multi-faceted user satisfaction [112]. Furthermore, Reinforcement Learning from Personalized Human Feedback (RLPHF) decomposes alignment into user-specific dimensions, utilizing parameter merging techniques to synthesize a unified model that adheres to individual preferences rather than aggregate societal norms [113]. This is complemented by methods like Personalized-RLHF, which jointly learns user representations and reward models to capture latent preference structures from heterogeneous feedback data [37].

To mitigate the reliance on expensive human annotations, self-generated and iterative preference learning strategies have gained traction. Approaches such as Iterative Length-Regularized DPO (iLR-DPO) leverage online preferences labeled by reward models to progressively enhance model quality while penalizing verbosity, a common failure mode in generative lists [114]. Similarly, Reinforced Self-Training (ReST) utilizes offline generated samples to refine policies, offering a sample-efficient alternative to online RLHF [115]. In scenarios where pairwise preferences are scarce, Direct Reward Optimization (DRO) enables alignment using single-trajectory feedback (e.g., thumbs-up/down), broadening the applicability of preference learning to abundant but noisy interaction logs [116].

Critically, aligning generative policies with ranking objectives requires bridging the gap between language generation losses and permutation-sensitive evaluation metrics. Softmax-DPO (S-DPO) integrates negative sampling into the DPO framework, effectively instilling ranking information into the language model to distinguish preferred items from hard negatives, thus addressing the likelihood decline issue observed in vanilla DPO [117]. Additionally, regularization techniques such as Regularized Preference Optimization (RPO) combine preference losses with supervised fine-tuning objectives to prevent overoptimization against imperfect reward signals, ensuring robustness in sparse data regimes [118]. Future research must further explore active query mechanisms to reduce annotation costs [119] and develop scalable routing strategies that dynamically select alignment policies based on user context [120], ultimately fostering generative recommenders that are not only accurate but also deeply aligned with the nuanced and evolving intentions of individual users.

5.2 Hallucination Mitigation and Factuality Assurance

Building upon the alignment strategies discussed previously, which optimize user preference fidelity, a critical challenge emerges: ensuring that the generated recommendations correspond to real, existing items rather than plausible fabrications. Hallucination in generative recommendation represents a fundamental deviation from traditional ranking paradigms, where Large Language Models (LLMs) may invent non-existent items, attributes, or user interactions, thereby compromising system reliability. This phenomenon stems from the inherent tendency of autoregressive models to prioritize semantic fluency over factual grounding, a challenge exacerbated when mapping continuous latent spaces to discrete item catalogs [121]. To mitigate this, recent architectural innovations focus on constraining the generation space through rigorous item indexing and retrieval-augmented mechanisms. A primary strategy involves the construction of LLM-compatible item identifiers that uniquely map tokens to valid corpus entries, preventing the synthesis of "ghost" items. Approaches such as sequential, collaborative, and hybrid indexing have demonstrated that semantically rich, platform-agnostic textual IDs can significantly reduce hallucination rates compared to arbitrary numeric mappings [23; 122]. Furthermore, grounding the LLM in the recommendation space via a bi-step paradigm—where the model first generates meaningful tokens and then retrieves corresponding actual items—ensures that outputs remain tethered to the existing item pool [25].

Beyond indexing, Retrieval-Augmented Generation (RAG) serves as a critical architectural intervention to inject external knowledge and verify factual consistency during inference. By decoupling retrieval from generation, systems can constrain the LLM's output to a pre-verified candidate set, effectively limiting the probability of hallucinating unavailable entities [123; 124]. In multimodal contexts, hallucination often manifests as a misalignment between visual inputs and textual descriptions; here, preference fine-tuning frameworks like POVID have shown efficacy by generating dispreferred data containing plausible hallucinations to explicitly train the model against such errors [125]. Complementing these preventive measures, detection mechanisms leveraging internal state auditing offer a proactive layer of assurance. Techniques such as Activation Patching (AtP) and the analysis of massive activations enable the localization of model components responsible for factual errors, allowing for the identification of anomalous patterns prior to output generation [126; 127]. Additionally, automated auditing via discrete optimization can systematically search for input-output pairs that induce hallucinatory behaviors, facilitating robust pre-deployment testing [128].

Despite these advances, trade-offs persist between generation diversity and factual strictness. Over-constraining decoding may lead to homogeneity, while unconstrained generation risks validity. Emerging solutions like Debiasing-Diversifying Decoding (D3) address amplification bias and homogeneity, suggesting that nuanced decoding strategies are essential for balancing accuracy with recommendation novelty [26]. Furthermore, the integration of causal reasoning allows models to distinguish between spurious correlations and genuine user preferences, reducing hallucinations driven by popularity bias or data artifacts [85; 80]. Future research must focus on dynamic, training-free calibration methods, such as attention calibration techniques, to enhance factuality without incurring the computational costs of full retraining [129]. Ultimately, while these mechanisms secure the factual integrity of recommendations, they address only one dimension of trustworthiness; as the focus shifts from the validity of generated content to the safety of the system itself, new challenges arise regarding external adversarial manipulation and the internal protection of sensitive user data.

5.3 Privacy-Preserving Architectures and Data Sovereignty

The integration of Large Language Models (LLMs) into generative recommendation systems introduces profound privacy challenges, as the capacity for semantic reasoning often correlates with the risk of memorizing sensitive user interaction histories. Addressing these vulnerabilities requires a paradigm shift from centralized data aggregation to distributed, privacy-preserving architectures that uphold data sovereignty while maintaining recommendation utility. Federated Learning (FL) has emerged as a foundational framework for this transition, enabling model training on decentralized client devices without exposing raw behavioral logs. However, standard federated averaging protocols often overlook the heterogeneous contribution of individual users, leading to suboptimal global models. To mitigate this, attentive aggregation mechanisms have been proposed, which weight client updates based on their relevance to the global objective, thereby enhancing convergence and robustness against gradient inversion attacks in language modeling tasks [130].

Complementing distributed training, rigorous differential privacy (DP) guarantees are essential to prevent the extraction of personally identifiable information (PII) from model parameters. The phenomenon of emergent memorization, where LLMs verbatim reproduce training sequences, poses a critical threat to user confidentiality [131]. Consequently, confidentially redacted training methods and user-level DP mechanisms are increasingly employed to inject calibrated noise during the optimization process, ensuring that the presence or absence of any single user's interaction history remains statistically indistinguishable. Furthermore, the risk extends beyond training to inference, where benign interaction sequences can be exploited to infer sensitive attributes such as income or location. Mitigation strategies involve sanitization techniques and cascade systems that filter latent representations before they reach the generative decoder, effectively limiting attribute leakage during remote querying.

A pivotal component of data sovereignty in generative recommendation is the capability for efficient machine unlearning, enabling the "right to be forgotten" without computationally prohibitive full retraining. Recent advancements demonstrate that specific data subsets, such as copyrighted texts or sensitive user histories, can be erased from LLMs through targeted fine-tuning on alternative labels generated by the model itself, effectively approximating the state of a model never trained on the target data [132]. This is particularly relevant for recommendation systems where user preferences are dynamic; parameter-efficient fine-tuning (PEFT) architectures, such as Low-Rank Adaptation (LoRA), facilitate this by isolating user-specific parameters. By partitioning adapters, systems can selectively revoke access or erase specific user memories while preserving the integrity of the foundational model [28].

Despite these advancements, significant trade-offs remain between privacy guarantees and recommendation personalization. Stronger privacy constraints often degrade the model's ability to capture nuanced, long-tail user preferences, leading to generic suggestions. Moreover, the rise of synthetic data generation for privacy preservation introduces the risk of "model collapse," where training on AI-generated data causes the distribution of user behaviors to narrow irreversibly [13]. Future research must therefore focus on developing adaptive privacy budgets that dynamically adjust based on data sensitivity and exploring neuro-symbolic approaches that separate factual item knowledge from private user trajectories, ensuring that generative recommenders remain both trustworthy and socially responsible.

5.4 Robustness, Security, and Adversarial Defense

While privacy concerns the internal protection of sensitive user data, security focuses on external adversarial manipulation targeting the robustness of generative recommendation systems. The integration of Large Language Models (LLMs) introduces a distinct attack surface that extends beyond traditional collaborative filtering vulnerabilities, necessitating a rigorous re-evaluation of system resilience. Unlike discriminative models that operate on latent vectors, generative recommenders process and produce natural language, rendering them susceptible to semantic manipulation and prompt-based exploits. A primary vector is prompt injection, where malicious actors craft inputs to bypass safety filters or inject biased items, effectively hijacking the recommendation logic. Recent surveys highlight that even safety-aligned models remain vulnerable to "jailbreak" attacks that exploit linguistic ambiguities to generate harmful or manipulated content [133]. In the recommendation context, this manifests as stealthy attacks where subtle alterations to item textual content during the testing phase can significantly boost exposure without degrading overall system performance, a phenomenon difficult to detect due to its semantic subtlety [134].

Furthermore, the reliance on user profiles and interaction histories as context creates opportunities for adversarial profile manipulation. Attackers can inject synthetic interaction sequences designed to shift the model's latent representation of a user or the global item space, a threat exacerbated by the model's tendency to memorize training data [135]. This memorization risk, previously identified as a privacy liability, also serves as a security vector where the susceptibility of these systems is not merely theoretical; empirical studies demonstrate that LLM-based recommenders can be coerced into amplifying specific providers or genres through carefully constructed system roles and prompt strategies, revealing inherent biases that can be weaponized [91]. The risk is further compounded by the potential for LLMs to be used as weapons for automated influence operations, where generative agents create convincing, misleading content to manipulate user preferences at scale [136].

Defensive strategies must therefore evolve from simple input sanitization to comprehensive resilience frameworks that parallel the privacy-preserving architectures discussed earlier. One promising direction involves adversarial training and red-teaming, where models are exposed to diverse attack scenarios to harden their decision boundaries. Techniques such as automatic auditing via discrete optimization can proactively uncover failure modes by searching for input-output pairs that match targeted malicious behaviors [128]. Additionally, the concept of "unlearning" offers a dual-purpose mechanism: just as it enables the "right to be forgotten" for privacy, it provides a rapid response to identified security breaches or copyright violations by removing undesirable behaviors or the influence of poisoned data without full retraining [132; 137]. However, a critical trade-off exists between robustness and utility; excessive constraints may degrade the generative flexibility that defines these systems. Emerging research suggests that preference optimization techniques, such as Direct Preference Optimization (DPO), can be adapted to penalize insecure outputs while maintaining alignment with user preferences [36].

Looking forward, the field requires standardized resilience evaluation frameworks that quantify robustness against black-box attacks and simulate complex, multi-step adversarial workflows. The intersection of cybersecurity and recommendation demands a holistic view where defense mechanisms are embedded within the architecture, leveraging tools for real-time threat detection and anomaly identification [138; 139]. Ultimately, ensuring the integrity of generative recommenders requires a shift from reactive patching to proactive, design-level security that accounts for the unique semantic vulnerabilities of language-driven personalization, laying the groundwork for the broader ethical governance challenges regarding fairness and autonomy discussed in the subsequent section.

5.5 Ethical Governance, Fairness, and Societal Impact

The deployment of generative Large Language Models (LLMs) in recommendation systems precipitates a paradigm shift in ethical governance, moving beyond traditional metric optimization to address profound societal implications regarding fairness, transparency, and agent autonomy. Unlike discriminative models that largely operate on latent collaborative signals, generative recommenders leverage open-world knowledge, thereby inheriting and potentially amplifying the social prejudices embedded within pre-training corpora. Recent empirical evaluations indicate that LLM-based recommenders exhibit significant disparities across sensitive user attributes, necessitating specialized fairness benchmarks like FaiRLLM to rigorously assess provider fairness and temporal stability [140]. Furthermore, item-side fairness remains a critical concern, as inherent semantic biases in LLMs can skew exposure opportunities for vulnerable content providers, requiring calibrated frameworks such as IFairLRS to mitigate these distortions [141].

A pivotal challenge in this domain is the detection and reflective debiasing of deep-seated biases that manifest not only in output distributions but also in the reasoning pathways of generative agents. While traditional methods rely on post-hoc statistical parity, emerging approaches utilize multi-model consensus and causal inference to uncover spurious correlations and distinguish genuine preferences from popularity bias [80; 142]. However, the efficacy of these debiasing strategies is often contingent on the quality of evaluation protocols, which frequently overlook the interplay between personalization and fairness, inadvertently perpetuating inequitable practices [143]. Moreover, the decoding process itself introduces unique ethical risks; standard length normalization can induce amplification bias, inflating scores for items with high-probability "ghost tokens" and leading to homogeneity that restricts information diversity [26].

Transparency and user controllability constitute the second pillar of ethical governance, particularly as generative systems evolve into autonomous agents capable of multi-turn negotiation and proactive influence. The "black-box" nature of LLM reasoning complicates the provision of faithful explanations, yet frameworks like LLMHG demonstrate that synergizing hypergraph learning with LLM reasoning can enhance human-centric explainability [73]. Crucially, users must retain agency over their digital personas; recent advances in learning latent preferences from user edits enable personalized steering without costly fine-tuning, fostering a collaborative alignment where users can inspect and modify the system's understanding of their intent [76]. Nevertheless, the rise of autonomous agents raises alarms regarding manipulative behaviors and the potential for "sycophancy," where models prioritize agreeable but potentially harmful responses to maximize reward signals [74].

Looking toward the long-term societal impact, the proliferation of generative agents threatens to exacerbate filter bubbles and induce "model collapse" if training pipelines increasingly rely on synthetic, LLM-generated data that lacks the diversity of genuine human interaction [13; 84]. The capacity of LLMs to simulate human discourse with high fidelity also blurs the line between organic and automated influence, posing risks for democratic processes and information ecosystems [144; 136]. Consequently, future research must prioritize the development of diversity-aware decoding strategies and robust governance frameworks that ensure generative recommenders promote a pluralistic information environment rather than reinforcing echo chambers. Ultimately, achieving trustworthy generative recommendation requires a holistic integration of causal fairness, interpretable agency, and strict safeguards against the erosion of data sovereignty and societal well-being.

6 Evaluation Protocols, Benchmarks, and Deployment Challenges

6.1 Holistic Evaluation Metrics Beyond Ranking Accuracy

The paradigm shift toward generative recommendation necessitates a fundamental re-evaluation of assessment protocols, moving beyond the sufficiency of traditional ranking metrics like Precision@K or NDCG. While these discriminative measures effectively gauge item relevance, they fail to capture the unique generative capacities of Large Language Models (LLMs), such as semantic coherence, explanatory fidelity, and the synthesis of novel content. Consequently, a holistic evaluation framework must integrate multidimensional criteria that align with the generative nature of these systems.

A primary dimension of this expanded framework is the assessment of semantic coherence and personalization depth. Unlike conventional models that output opaque IDs, generative recommenders produce natural language rationales or item descriptions that must align logically with user history. Automated metrics leveraging LLMs as evaluators, such as those proposed in [145], offer a scalable alternative to costly human judgment by correlating model outputs with high-quality human preferences. Furthermore, [146] establishes benchmarks specifically designed to evaluate personalized text generation, highlighting that standard accuracy metrics often overlook the nuance of individual user alignment. The challenge lies in quantifying "personalization" beyond lexical overlap; recent approaches suggest evaluating the consistency of generated profiles against latent user intents, as seen in [147], where the quality of the generated user summary directly impacts downstream predictive performance.

Equally critical is the evaluation of diversity and serendipity, areas where generative models face distinct risks of homogeneity. The autoregressive nature of LLMs can lead to "amplification bias," where high-probability tokens dominate outputs, resulting in repetitive recommendations. [26] identifies this phenomenon and proposes decoding strategies to counteract it, suggesting that evaluation metrics must explicitly penalize syntactic and semantic repetition. Traditional diversity metrics based on item categories are insufficient for generative settings; instead, metrics must assess the novelty of generated narratives and the structural variety of reasoning paths. [148] provides a rigorous method for measuring the novelty of generated text relative to training corpora, a technique adaptable for ensuring recommendations do not merely regurgitate popular items but explore the long tail effectively.

The trustworthiness of generative recommendations hinges on the quality and faithfulness of explanations. [73] emphasizes that explainability is not merely an add-on but a core component of user-centric modeling. Evaluation here requires measuring the logical consistency between the generated rationale and the actual user-item interaction, ensuring the model does not hallucinate non-existent features or preferences. [149] warns that without robust grounding metrics, generative systems risk producing persuasive but factually incorrect justifications. Moreover, fairness evaluations must evolve to account for personalization; [143] argues that current frameworks often ignore personalization factors, inadvertently masking biased outcomes under the guise of tailored experiences.

Ultimately, the future of evaluation in generative recommendation lies in the synthesis of automated, LLM-based judging with human-centric validation. As [150] suggests, there is often a misalignment between model optimization objectives and human expectations of generalization. Therefore, next-generation metrics must bridge this gap by incorporating feedback loops that assess not only the static quality of a recommendation list but also the dynamic impact of generative interactions on long-term user satisfaction and ecosystem health.

6.2 Human-Centric and Agent-Based Simulation Protocols

Building upon the limitations of static, content-centric metrics identified previously, the evaluation landscape must evolve toward dynamic, human-centric protocols that capture the interactive and longitudinal nature of generative recommendation. While static assessments measure immediate semantic coherence and diversity, they fail to account for the multi-turn dialogue capabilities of Large Language Models (LLMs) or the long-term societal impacts of algorithmic curation. To bridge this gap, contemporary research increasingly leverages LLM-powered agents as high-fidelity user simulators, enabling scalable, controlled, and ethically safe experimentation within complex socio-technical environments. This shift from offline log analysis to agent-based simulation provides the necessary infrastructure to test systems against emergent social dynamics that traditional datasets cannot reveal.

A primary paradigm in this domain is the deployment of generative agents equipped with distinct personas, memory modules, and reflection mechanisms to mimic human decision-making processes. Frameworks such as [81] demonstrate the capacity of LLM-empowered agents to simulate not only taste-driven actions but also emotion-driven behaviors, providing a sandbox for studying phenomena like filter bubbles and causal relationships in recommendation tasks. Similarly, [151] extends this capability to web search, generating unique user profiles at scale to simulate diverse search behaviors, thereby overcoming the scarcity and privacy constraints of real-world data. These agent-based simulators allow researchers to conduct stress tests on recommendation policies, evaluating long-term engagement and the robustness of systems against emergent social dynamics. Furthermore, [152] highlights the importance of coordinating multiple agents to simulate task-oriented social interactions, revealing both the promise and current limitations of LLMs in modeling complex human coordination.

Beyond unilateral simulation, advanced protocols emphasize interactive and collaborative evaluation to deepen the realism of these environments. The [153] framework introduces a novel approach where both users and items are modeled as autonomous agents that interact and reflect on disparities between simulated and real-world records, effectively internalizing collaborative filtering signals within an agent-based loop. This bidirectional interaction enhances the modeling of non-verbal behaviors, such as item clicking, which pure language modeling often overlooks. In conversational settings, protocols like [154] utilize LLM-based user simulators to evaluate the interactive nature of Conversational Recommender Systems (CRS), moving beyond ground-truth matching to assess intent clarification and negotiation effectiveness. Complementary work in [101] establishes rigorous multi-task protocols to measure how well synthetic users emulate human properties, including open-ended preference expression and feedback provision, ensuring that simulators do not merely replicate surface-level statistics but capture underlying cognitive processes.

Despite these advancements, significant challenges remain regarding the fidelity and artifactuality of simulated behaviors, which directly impacts the reliability of the insights derived from these dynamic evaluations. Studies such as [84] warn of hidden disparities between LLM-generated and human-generated data, particularly in complex tasks requiring nuanced understanding. Moreover, [155] reveals that LLMs may exhibit inconsistent personality traits across different scenarios, fundamentally differing from human psychological consistency. Consequently, future evaluation protocols must integrate human-in-the-loop mechanisms, as seen in [156], where LLMs guide human workers to create realistic, personalized dialogue data, bridging the gap between fully synthetic efficiency and human authenticity. Ultimately, the field must strive for hybrid evaluation ecosystems that combine the scalability of agent-based sandboxes with the grounding of human judgment. However, even as these simulation frameworks mature, their practical adoption in industrial settings faces a formidable barrier: the severe computational cost of running large-scale, multi-turn agent interactions in real-time. This tension between the need for high-fidelity dynamic evaluation and the constraints of inference efficiency necessitates a deep dive into optimization strategies, forming the critical focus of the subsequent discussion.

6.3 Efficiency Optimization and Scalable Deployment Architectures

The deployment of Large Language Models (LLMs) for generative recommendation in industrial settings necessitates a fundamental re-engineering of inference pipelines to address the prohibitive latency and memory costs associated with autoregressive generation. The primary bottleneck lies in the quadratic complexity of attention mechanisms and the linear growth of the Key-Value (KV) cache with sequence length, which severely constrains throughput in high-traffic environments. To mitigate this, recent advancements have pivoted towards speculative decoding strategies. [157] proposes restructuring speculative batches into trees to maximize token acceptance rates, while [158] introduces a tree-based parallel verification mechanism that accelerates serving by up to 3.5x without compromising generative quality. Furthermore, [87] demonstrates that instruct-tuning models to perform auto-parallel generation can reduce decoding steps by half, offering a complementary algorithmic speedup.

Memory management remains equally critical, particularly for modeling long user behavior sequences where the KV cache often exceeds GPU capacity. Innovative eviction policies have emerged to address this; [159] identifies that a sparse set of "heavy hitter" tokens contributes disproportionately to attention scores, proposing a dynamic eviction policy that retains these critical tokens alongside recent context to improve throughput by up to 29x. Complementing this, [160] achieves near-lossless compression by integrating ultra-low precision quantization with low-rank approximation and sparse error correction, effectively transforming inference from a memory-bound to a compute-bound problem. Additionally, [161] showcases architectural innovations like Multi-head Latent Attention (MLA), which compresses the KV cache into latent vectors, reducing cache memory usage by over 93% while maintaining state-of-the-art performance.

Beyond algorithmic optimizations, scalable deployment increasingly relies on heterogeneous computing and model routing. [162] demonstrates that by aggregating memory across GPU, CPU, and disk via optimized linear programming, massive models can be served on commodity hardware with high throughput. For cost-performance balancing, [120] introduces learned routers that dynamically dispatch requests between powerful and efficient models based on query complexity, significantly reducing operational costs. In the specific context of sequential recommendation, [86] argues that standard beam search is often unnecessary, proposing Lite-LLM4Rec which utilizes a direct item projection head to achieve a 97% efficiency improvement. Similarly, [58] leverages selective state space models to replace attention mechanisms entirely, offering linear complexity scaling for long user histories.

Looking forward, the convergence of these techniques suggests a shift towards hybrid architectures that combine sparse activation, aggressive cache compression, and adaptive decoding. However, challenges persist in maintaining recommendation diversity and personalization fidelity under extreme compression. Future research must explore the interplay between efficient inference and the preservation of nuanced user preferences, ensuring that latency optimizations do not degrade the semantic coherence essential for generative recommendation.

6.4 Benchmark Datasets, Standardization, and Security Robustness

While robust evaluation protocols and secure benchmarking frameworks are essential for validating generative recommenders, their practical realization in industrial settings hinges on overcoming the severe computational bottlenecks of autoregressive generation. The deployment of Large Language Models (LLMs) for generative recommendation necessitates a fundamental re-engineering of inference pipelines to address the prohibitive latency and memory costs that often contradict the real-time demands of high-traffic environments. The primary bottleneck lies in the quadratic complexity of attention mechanisms and the linear growth of the Key-Value (KV) cache with sequence length, which severely constrains throughput. To mitigate this, recent advancements have pivoted towards speculative decoding strategies. [157] proposes restructuring speculative batches into trees to maximize token acceptance rates, while [158] introduces a tree-based parallel verification mechanism that accelerates serving by up to 3.5x without compromising generative quality. Furthermore, [87] demonstrates that instruct-tuning models to perform auto-parallel generation can reduce decoding steps by half, offering a complementary algorithmic speedup.

Memory management remains equally critical, particularly for modeling long user behavior sequences where the KV cache often exceeds GPU capacity—a challenge directly impacting the fidelity of the user simulations discussed previously. Innovative eviction policies have emerged to address this; [159] identifies that a sparse set of “heavy hitter” tokens contributes disproportionately to attention scores, proposing a dynamic eviction policy that retains these critical tokens alongside recent context to improve throughput by up to 29x. Complementing this, [160] achieves near-lossless compression by integrating ultra-low precision quantization with low-rank approximation and sparse error correction, effectively transforming inference from a memory-bound to a compute-bound problem. Additionally, [161] showcases architectural innovations like Multi-head Latent Attention (MLA), which compresses the KV cache into latent vectors, reducing cache memory usage by over 93% while maintaining state-of-the-art performance.

Beyond algorithmic optimizations, scalable deployment increasingly relies on heterogeneous computing and model routing to balance the cost-performance trade-offs inherent in large-scale benchmarking. [162] demonstrates that by aggregating memory across GPU, CPU, and disk via optimized linear programming, massive models can be served on commodity hardware with high throughput. For cost-performance balancing, [120] introduces learned routers that dynamically dispatch requests between powerful and efficient models based on query complexity, significantly reducing operational costs. In the specific context of sequential recommendation, [86] argues that standard beam search is often unnecessary, proposing Lite-LLM4Rec which utilizes a direct item projection head to achieve a 97% efficiency improvement. Similarly, [58] leverages selective state space models to replace attention mechanisms entirely, offering linear complexity scaling for long user histories.

Looking forward, the convergence of these techniques suggests a shift towards hybrid architectures that combine sparse activation, aggressive cache compression, and adaptive decoding. However, challenges persist in maintaining recommendation diversity and personalization fidelity under extreme compression, a concern that echoes the warnings about data artifacts and model collapse in the subsequent discussion on benchmarking. Future research must explore the interplay between efficient inference and the preservation of nuanced user preferences, ensuring that latency optimizations do not degrade the semantic coherence essential for generative recommendation or compromise the security and privacy guarantees required for trustworthy deployment.

The evaluation of Generative Recommendation systems necessitates a rigorous synthesis of standardized benchmarking frameworks and robust security protocols, as the paradigm shift from discriminative ranking to autoregressive generation introduces unique vulnerabilities and data requirements. Currently, the landscape of public datasets remains fragmented, often lacking the semantic richness required to fully exploit Large Language Model (LLM) capabilities. While platforms like OpenP5 strive to unify data processing and evaluation pipelines, there is an urgent need for benchmarks that explicitly account for long-tailed distributions and complex user-item structural properties [163]. To address data sparsity and cold-start challenges, recent approaches leverage LLMs to generate synthetic training data; however, the efficacy of such synthesized data is highly contingent on the subjectivity of the classification task, with performance degrading in highly subjective domains [164]. Furthermore, the reliance on model-generated content risks "model collapse," where the diversity of the data distribution irreversibly diminishes over iterative training cycles, underscoring the critical value of authentic human interaction logs [13]. Consequently, future standardization efforts must prioritize datasets that facilitate the evaluation of pluralistic alignment, ensuring models can cater to diverse user values rather than reinforcing majority biases [165].

Concurrently, the integration of LLMs into recommendation pipelines expands the attack surface, introducing sophisticated security threats that transcend traditional adversarial manipulation. A distinct vulnerability in generative recommenders is the susceptibility to stealthy text-based attacks, where adversaries subtly alter item descriptions to manipulate exposure without triggering standard detection mechanisms or degrading overall system performance [134]. This contrasts with conventional attacks on collaborative filtering, as it exploits the semantic understanding inherent in the generative backbone. Moreover, the deployment of LLMs raises severe privacy concerns regarding memorization, where models may inadvertently regurgitate sensitive user interaction histories or personally identifiable information (PII) verbatim [131]. Detection of such leakage has advanced through methods like Relative Conditional Log-Likelihoods (ReCaLL), which identify membership inference risks by analyzing shifts in conditional probabilities when non-member contexts are introduced [166]. Additionally, the "goldfish loss" mechanism has been proposed as a defensive training objective to mitigate verbatim memorization by excluding random tokens from loss computation, thereby preserving utility while enhancing privacy [135].

The intersection of benchmarking and security further reveals gaps in evaluating robustness against prompt injection and jailbreaking, which can compel models to bypass safety filters or promote biased items [133]. Effective defense requires not only constrained decoding but also comprehensive risk taxonomies that quantify resilience against black-box attacks targeting the generative process [167]. As the field matures, the development of unified evaluation standards must integrate these security dimensions, moving beyond accuracy metrics to include measures of privacy leakage, resistance to adversarial perturbations, and the stability of synthetic data ecosystems. Only through such holistic benchmarking can the community ensure the trustworthy deployment of generative recommendation systems at scale.

7 Conclusion

The integration of Large Language Models (LLMs) into recommendation systems marks a definitive paradigm shift from discriminative ranking to generative synthesis, fundamentally redefining how user behaviors are modeled and items are retrieved. This survey has elucidated that moving beyond traditional ID-based collaborative filtering allows for the unification of semantic understanding and interaction dynamics. Unlike conventional deep learning architectures that treat items as opaque indices [168], generative approaches leverage the extensive world knowledge of LLMs to interpret user intent through natural language instructions [88]. By framing recommendation as a text-to-text generation task, systems can now directly generate item identifiers or descriptive sequences, effectively addressing cold-start problems and data sparsity that have long plagued discriminative models [169; 170].

However, this transition introduces critical challenges regarding architectural efficiency and factual consistency. While autoregressive generation offers superior interpretability and flexibility [171], it incurs significant latency compared to parallel scoring mechanisms. Recent advances in parameter-efficient adaptation, such as user-specific LoRA modules [28] and retrieval-augmented generation [99], attempt to mitigate these costs, yet the trade-off between reasoning depth and inference speed remains a primary bottleneck. Furthermore, the phenomenon of "model collapse," where training on synthetic data degrades distributional fidelity [13], poses an existential threat to the long-term viability of generative recommenders that rely on self-generated feedback loops. Ensuring that generated recommendations correspond to valid, existing items requires rigorous constrained decoding and verification protocols to prevent hallucination [163].

Looking forward, the frontier of generative recommendation lies in the development of autonomous, self-evolving agents capable of lifelong learning without catastrophic forgetting [34]. The integration of neuro-symbolic reasoning promises to combine the structural precision of graph neural networks with the semantic fluency of LLMs, enabling robust causal inference and counterfactual simulation. Moreover, as multimodal capabilities expand, future systems must seamlessly synthesize text, vision, and audio to model complex user preferences in heterogeneous environments [121; 172]. Ultimately, the trajectory of this field depends on establishing rigorous evaluation benchmarks that assess not only ranking accuracy but also the ethical alignment, diversity, and societal impact of generative policies. Only by addressing these open challenges can we realize the full potential of LLMs to create truly personalized, transparent, and resilient recommendation ecosystems.

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